WORKED SOLUTION

STEP II 2003, Question 9

Mechanics • Limiting Friction Equilibrium

STEP II 2003, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q9

The key idea

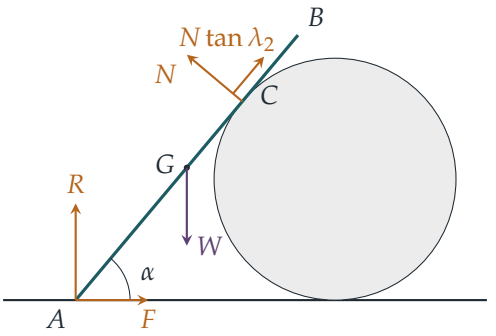
Because the coefficients of friction are given as $\tan\lambda_1$ and $\tan\lambda_2$, the natural tool is the *total reaction* (normal reaction plus friction combined into one force). When friction is limiting, the total reaction makes the angle of friction λ with the normal. That leaves the rod under just three forces whose directions we know, and resolving horizontally and vertically gives two equations in two unknown magnitudes. The only real work is deciding which way each friction force points.

Solution

The model and the directions of friction

We treat the rod as a rigid uniform rod and the cylinder as fixed. The rod touches the cylinder along a tangent, so the normal reaction at C is perpendicular to the rod. Take the cylinder to lie on the side of A in the positive horizontal direction. The forces on the rod are then:

- its weight W, acting vertically down at the midpoint G;
- at A: a normal reaction R vertically up and a friction force F along the floor;
- at C: a normal reaction N perpendicular to the rod, pointing up and back towards A, together with a friction force along the rod.



Ask yourself Which way does friction act at A and at C? Could the rod be on the point of slipping the other way?

The rod has only one way to move while keeping both contacts: as A slides along the floor, the angle α changes and the rod slides over the cylinder. If A moves away from the cylinder, the rod drops (so α decreases) and the rod's point at C slides *down* along the rod relative to the cylinder. If A moves towards the cylinder, everything is reversed.

In the second case friction at A would point away from the cylinder and friction at C would point down the rod. Then all three horizontal forces (friction at A, the horizontal part of N, and the horizontal part of friction at C) would point away from the cylinder, and horizontal equilibrium would be impossible. So in limiting equilibrium the rod is about to slide down: friction on the rod at A acts *towards* the cylinder, and friction at C acts *up* the rod, as drawn.

Total reactions

Let S be the total reaction at A and T the total reaction at C. Since friction is limiting at both contacts:

- S makes angle λ_1 with the upward vertical, tilted towards the cylinder, because $F = R\tan\lambda_1$.
- The normal at C makes angle α with the upward vertical, tilted back towards A, because the rod makes angle α with the horizontal. Friction up the rod turns T by λ_2 back towards the vertical. So T makes angle $\alpha - \lambda_2$ with the upward vertical, measured on the side away from the cylinder.

Resolving for the rod,

$$\begin{aligned} \text{horizontally:} \quad & S\sin\lambda_1 = T\sin(\alpha-\lambda_2), \\ \text{vertically:} \quad & S\cos\lambda_1 + T\cos(\alpha-\lambda_2) = W. \end{aligned}$$

To eliminate T, we multiply the first equation by $\cos(\alpha - \lambda_2)$, the second by $\sin(\alpha - \lambda_2)$, and subtract the first from the second:

$$\begin{aligned} S\left[\cos\lambda_1 \sin(\alpha - \lambda_2) + \sin\lambda_1 \cos(\alpha - \lambda_2)\right] &= W\sin(\alpha - \lambda_2), \\ S\sin(\alpha + \lambda_1 - \lambda_2) &= W\sin(\alpha - \lambda_2), \end{aligned}$$

using the addition formula for sin. We are told that $\alpha \neq \lambda_2 - \lambda_1$, so $\sin(\alpha + \lambda_1 - \lambda_2) \neq 0$ (all the angles are acute, so $\alpha + \lambda_1 - \lambda_2$ lies strictly between $-\frac{\pi}{2}$ and π). Hence we may divide:

$$S = \frac{W\sin(\alpha - \lambda_2)}{\sin(\alpha + \lambda_1 - \lambda_2)}.$$

The frictional force at A is the horizontal component of S, namely $F = S\sin\lambda_1$, so

$$F = \frac{W\sin\lambda_1\ \sin(\alpha - \lambda_2)}{\sin(\alpha + \lambda_1 - \lambda_2)},$$

as required. In the same way, $T = W\sin\lambda_1 / \sin(\alpha + \lambda_1 - \lambda_2)$.

Ask yourself We never took moments. Isn't that suspicious?

Not here. Resolving in two directions is enough to find the two unknown magnitudes. Taking moments about A (where S has no moment, and nor does the friction at C, which acts along the rod) gives $N \cdot AC = W \cdot AG\cos\alpha$ with $N = T\cos\lambda_2$. This only tells us *where* C must be for limiting equilibrium:

$$\frac{AC}{AB} = \frac{\cos\alpha\ \sin(\alpha + \lambda_1 - \lambda_2)}{2\sin\lambda_1\cos\lambda_2}.$$

The given condition $AC > CB$ isn't needed for the friction force. Physically we also need $S > 0$ and $T > 0$, which holds exactly when $\alpha > \lambda_2$: a rod resting on the cylinder at a shallower angle than λ_2 can't be on the point of sliding down it.

Checks

- Smooth floor, $\lambda_1 = 0$: the formula gives $F = 0$, as it must.
- $\alpha = \lambda_2$: then T is vertical, it alone holds up the rod, and $F = 0$. This is the borderline angle at which the rod can just rest on the cylinder.
- Numerically, with $\lambda_1 = 0.3$, $\lambda_2 = 0.5$, $\alpha = 1.0$ (radians), solving the component equations for R and N directly gives $F \approx 0.1975W$, which matches the formula. In this case moments about A then put C at about 0.75 of the way along the rod, consistent with $AC > CB$.

Common slips

- Taking the reaction at C to be vertical, or perpendicular to the cylinder's "top". The reaction is normal to the common tangent, which is the rod itself.

- Choosing the friction directions without checking them. The wrong pair makes horizontal equilibrium impossible.
- Tilting the total reaction at C the wrong way, which gives $\alpha + \lambda_2$ in place of $\alpha - \lambda_2$.

Beyond STEP

Taking the total reaction at angle λ to the normal is the idea behind the *cone of friction*: a contact can hold as long as the total reaction stays inside a cone of half-angle λ around the normal. With three forces, the lines of action must also pass through one point, which turns many statics problems into geometry. Problems with several rough contacts, where the order in which they slip matters, are a good introduction to rigid-body statics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body equilibrium carefully.

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